The Monetary Policy and Aggregate Demand Curves

Chapter 21

Econ 351 – Monetary Economics

University of Tennessee, Knoxville

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Preview and Learning Objectives

This chapter covers three ideas:

- why central banks raise interest rates when inflation rises,
- how this gives us the monetary policy curve,
- how the monetary policy curve and the *IS* schedule together determine the aggregate demand curve.

Outline

The Federal Reserve and Monetary Policy

Monetary Policy (MP) Curve

Aggregate Demand (AD) Curve

The Federal Reserve and Monetary Policy

Role of the Federal Reserve

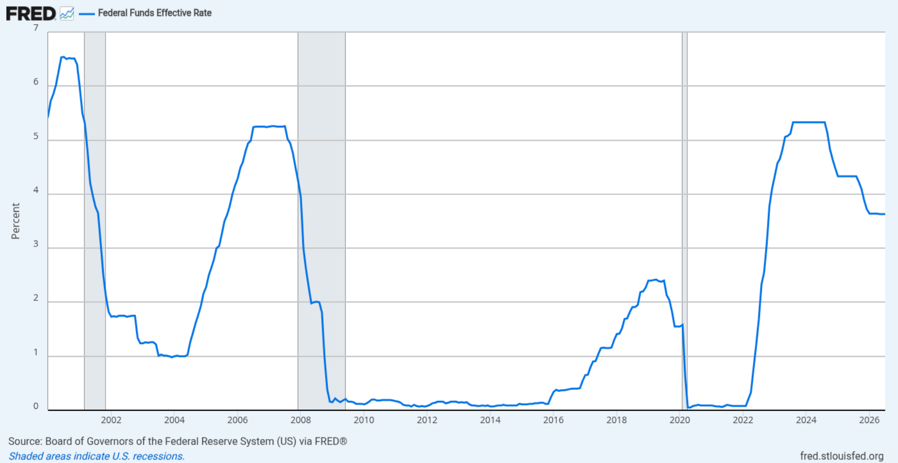
The Federal Reserve conducts monetary policy by setting the **federal funds rate**.

- The federal funds rate is the interest rate at which banks lend to each other.
- When the Fed **lowers** the federal funds rate, real interest rates fall.
- When the Fed **raises** the federal funds rate, real interest rates rise.

The Federal Open Market Committee, which sets the federal funds rate target, meeting in Washington, April 2016.

Wikimedia Commons, Federal Reserve, Public domain

The Federal Funds Rate Since 2000



Effective federal funds rate, percent: cuts in the 2001 and 2008–09 recessions and in 2020, hikes in 2004–06 and 2022–23.

Source: FRED, Federal Reserve Bank of St. Louis

Outline

The Federal Reserve and Monetary Policy

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The Monetary Policy (MP) Curve

Monetary Policy Rule

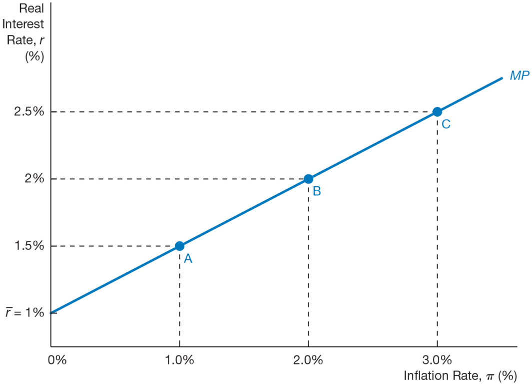
The MP curve describes how the central bank sets the real interest rate in response to inflation.

$$r = \bar{r} + \lambda\pi$$

- $\bar{r} =$ autonomous component of the real interest rate
- $\lambda =$ responsiveness of interest rates to inflation

The MP curve is **upward sloping**

Higher inflation $\Rightarrow$ higher real interest rates.



The monetary policy curve: the real interest rate the central bank sets at each inflation rate.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

The Taylor Principle

Why the *MP* Curve Slopes Upward

Central banks raise interest rates when inflation rises in order to stabilize inflation.

Taylor Principle

When inflation increases, the central bank raises the **nominal interest rate more than one-for-one**.

$$\pi \uparrow \quad \Rightarrow \quad r \uparrow$$

Higher interest rates reduce spending and stabilize inflation.

> Raising the nominal rate by more than the rise in inflation is what makes the *real* interest rate rise; only then does higher inflation reduce spending.

Shifts in the *MP* Curve

Two Types of Monetary Policy Changes

- **Automatic changes** (Taylor principle)
  - Movements **along** the *MP* curve
- **Autonomous policy changes**
  - Shift the MP curve

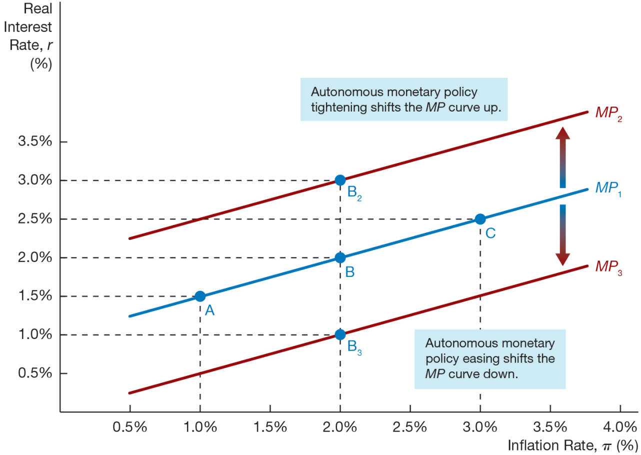
Examples

- Monetary tightening $\Rightarrow$ MP shifts up
- Monetary easing $\Rightarrow$ MP shifts down

Fed Chair Jerome Powell announcing a policy decision at an FOMC press conference: an autonomous tightening or easing shifts the MP curve.

Wikimedia Commons, Federal Reserve, Public domain

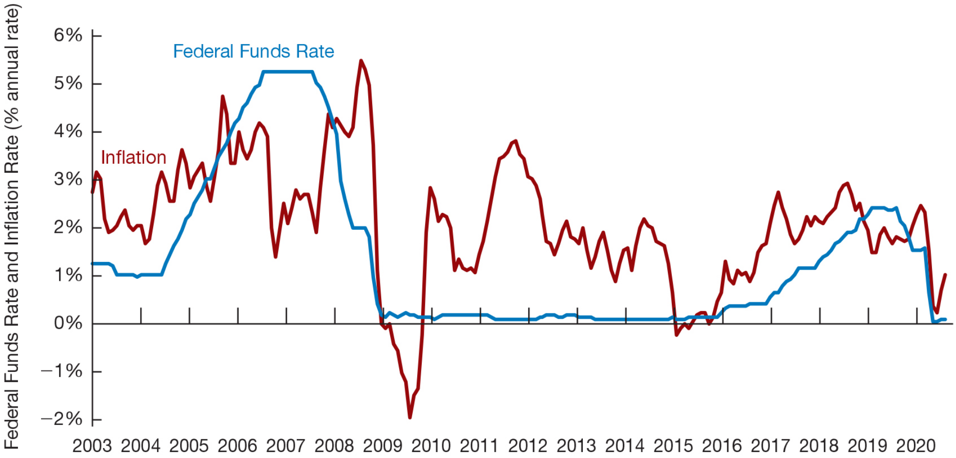
Shifts in the Monetary Policy Curve



Autonomous tightening shifts the MP curve up; autonomous easing shifts it down.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

The Federal Funds Rate and Inflation Rate, 2003–2020



The federal funds rate has tended to rise with inflation, as the MP curve implies.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Outline

The Federal Reserve and Monetary Policy

Monetary Policy (MP) Curve

Aggregate Demand (AD) Curve

The Aggregate Demand Curve

The aggregate demand (AD) curve shows the relationship between

- the inflation rate
- aggregate output

when the goods market is in equilibrium.

The AD curve has a **downward slope**.

Higher inflation $\Rightarrow$ higher interest rates $\Rightarrow$ lower aggregate demand.

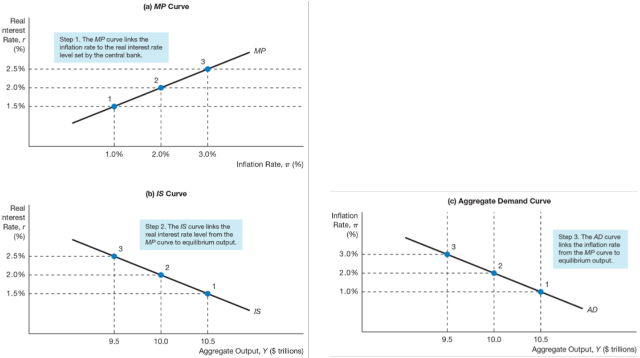
The AD curve is derived from:

- the **MP curve**, and
- the **IS curve**.

Deriving the AD Curve

Mechanism

- Inflation rises
- Central bank raises interest rates
- Investment and spending fall
- Output declines



Deriving the AD curve: each inflation rate maps through the MP curve to a real interest rate and through the IS curve to output.

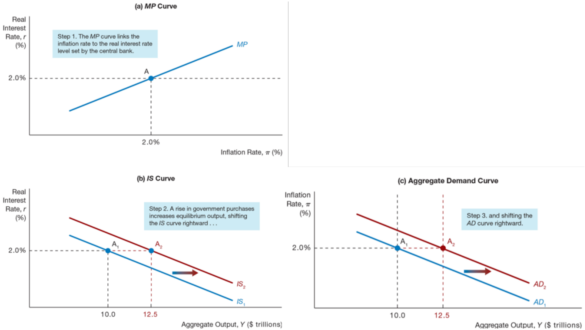
Source: Mishkin, The Economics of Money, Banking and Financial Markets

Factors That Shift the AD Curve

Shifts in the IS Curve

- Autonomous consumption
- Autonomous investment
- Government purchases
- Taxes
- Autonomous net exports
- Financial frictions

Any factor that shifts the **IS curve** shifts the **AD curve in the same direction**.



Shift in the AD curve from shifts in the IS curve.

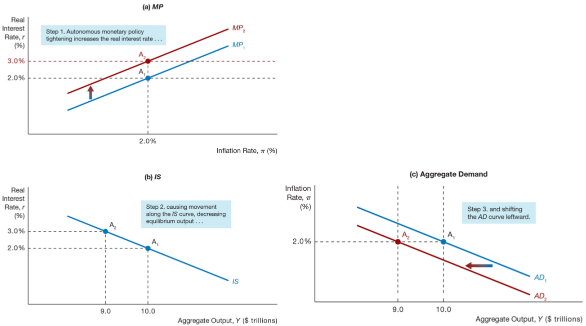
Source: Mishkin, The Economics of Money, Banking and Financial Markets

Shifts in the AD Curve from Monetary Policy

Shifts in the MP Curve

Monetary policy changes can also shift the AD curve.

- **Monetary tightening**
  - Higher real interest rates
  - Lower aggregate demand
  - AD shifts left
- **Monetary easing**
  - Lower real interest rates
  - Higher aggregate demand
  - AD shifts right



Shift in the AD curve from an autonomous monetary policy tightening.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Chapter summary

- The Fed conducts monetary policy by setting the federal funds rate; changes in it move real interest rates in the same direction.
- The MP curve, $r = \bar{r} + \lambda\pi$, is upward sloping: following the Taylor principle, the central bank raises nominal rates more than one-for-one with inflation so that the real rate rises.
- Automatic responses to inflation are movements along the MP curve; autonomous tightening or easing shifts it up or down.
- The AD curve combines the MP and IS curves: higher inflation raises real rates, lowers investment and net exports, and reduces output, so AD slopes downward.
- Anything that shifts the IS curve shifts AD in the same direction; autonomous monetary tightening shifts AD left and easing shifts it right.